Project Report

CS5100 Spring 2026

By

Siddhi Sanjay More

Introduction:

The management of invasive aquatic species like water hyacinth poses significant ecological challenges, requiring efficient and autonomous removal strategies. This project proposes a simulation based autonomous robotic system that uses learning by trying approaches to navigate a pond, map the environment, avoid obstacles and clean the water hyacinth. This project is inspired from my Undergraduate Hackathon Project **"SMART INDIA HACKATHON 2023**" which was motivated to innovate sustainable and environment friendly resources through technology. In 2023, this project was accomplished with the help of a team which designed mechanical and electrical systems of an autonomous mobile robot that cleaned the water bodies and used the hyacinth plants to create fiber used for clothing. As this project was classified as an Innovation strategy, no proper references or citations were possible.

Method:

The system architecture consists of a 3 -way Python framework which focuses on Environment (Environment.py), Q Learning Algorithm (Qlearning.py) and Main training and results (Main.py). The simulation environment consists of a 25×25 grid containing 30 obstacles and 20 randomly placed hyacinth plants.

- Environment- Consists of basic logic of the algorithm and rewards system
- Qlearning- This focuses on the agent's decision making
- Main- For training loop and visualization

The training duration was initially set to 1,000 episodes but then set to 5,000 episodes to ensure convergence with an increased learning rate of 0.2 and discount factor 0.95. The exploration strategy utilizes epsilon-greedy action selection, starting at epsilon = 1.0 and decaying at a rate of 0.995, allowing the agent to transition to exploitation around episode 600. Initially, randomizing the map every episode caused Q-table degradation as the agent encountered contradictory state-action values. The final approach maintains a stable obstacle layout, reshuffling only every 500 episodes to balance stable learning with generalizability.

Reward System Design:

Revisiting Penalty	-2
Completion	+100
Exploration	A coverage-scaled exploration bonus for visiting new cells.

Table 1 : Reward Table

When no targets (hyacinth) are available, the agent uses a 4x4 position zone, providing 16 distinct blind navigation policies with $O(1)$ computational efficiency.

Results:

The optimized agent demonstrated significant performance gains compared to the initial baseline. By episode 100, the agent achieved a 44% completion rate, which surged to 96% by episode 200. From episode 400 onward, a 100% completion rate was sustained, ensuring positive rewards to the agent indicating the removal of hyacinth plants exceeded the operational costs and penalties.

The adjustments and updates from final progress report to the final submission are as follows:

Parameters	Progress Report	Final Submission
Training Episodes	1,000	5,000
Max steps	1,000	2,000
Learning rate	0.1	0.2
Map type	15x15 (fixed state)	20x20 (randomized)

Table 2 : Parameters Comparison between Progress Reports stage and Submission Stage

Analysis:

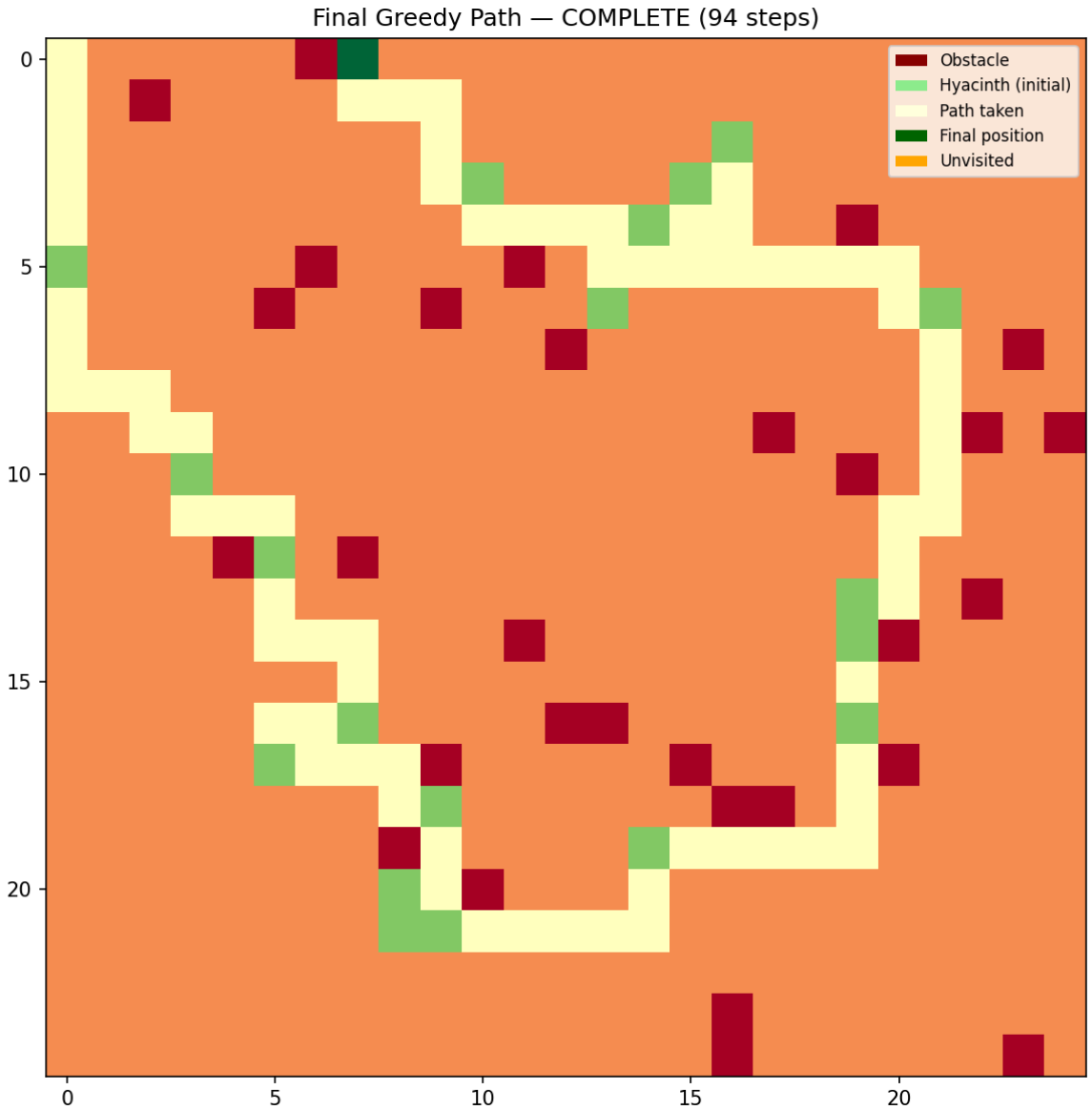

Figure 1: Final Greedy Path

This visualization represents the fully deterministic evaluation of the agent after training is complete. The "Path taken" (off-white line) shows the agent moving efficiently between targets while successfully navigating around the dark red obstacles. The path demonstrates that the agent's 4x4 position zones and vision radius allowed it to clear the entire grid without getting stuck in local loops.

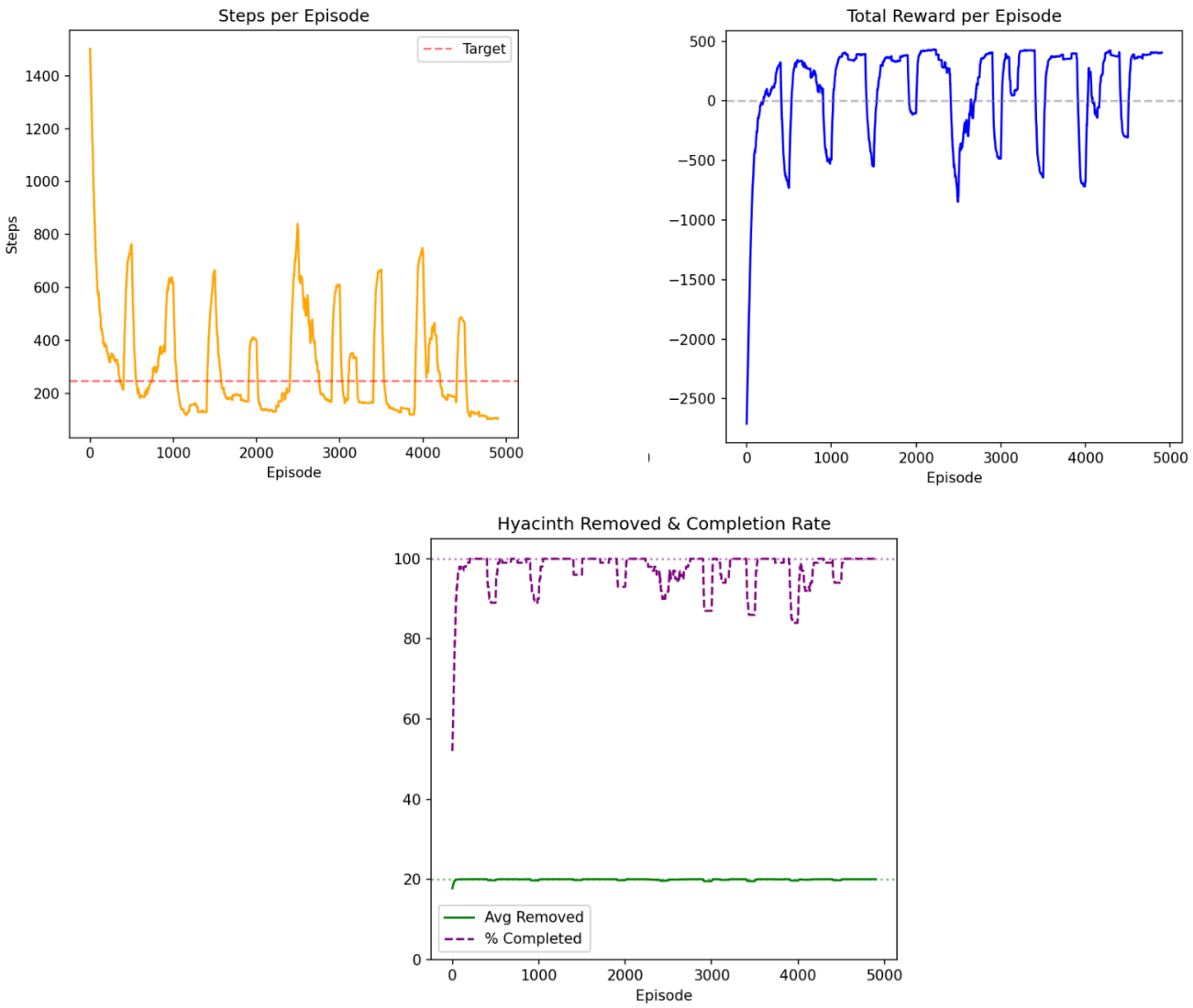

Figure 2: Subplots of Learning Results

Steps per Episode (Left): When the agent stops walking aimlessly, there is a significant initial decline in steps.The map's rearrangement, which requires the agent to momentarily re-learn the new obstacle layout, is represented by the periodic "spikes" that happen every 500 episodes.

Total Reward per Episode (Right): Rewards gradually increase toward positive values after beginning extremely negatively. The Q-table has converged when the reward curve stabilizes close to the top.

Hyacinth Removed & Completion Rate (Middle): As early as episode 400, the agent maintains a 100% completion rate.The agent's "Average Removed" (green line) remains nearly constant at 20 even when the map is rearranged, demonstrating that it has acquired a generalized method for locating targets even in novel configurations.

Key Improvements:

A robot in a physical aquatic environment can only "see" what its sensors can detect within a certain range; it cannot "see" the entire pond at once. For field robots, the prior version's reliance on global knowledge is impractical. The agent was unable to distinguish between being blind in the top-left and bottom-right regions of the pond since the previous technique reduced all "blind" scenarios to only four states.

To tackle this issue, I implemented a 4x4 grid of position zones providing 26 distinct blind navigation policies to guide the agent when targets are out of sight and reach. I also added a coverage scale exploration bonus to the agent for visiting unmapped cells, which is inspired by the SLAM robotic algorithm discovery phases.

This resulted in the agent transitioning from simple path following to an active search and clear strategy. Despite limited vision, the agent cleared all 20 hyacinths in 94 steps in the final evaluation. This architecture supports **MULTI ROBOT COORDINATIONS,** as agents can share a common “visited cells” to divide the space search without redundancy.

Github Link: https://github.com/more-sidd/hyacinth-cleaner-bot.git